Magnetic contact area P_{bs} and the angles Q_α, Q_β, Q_γ of unstructured B2.5 wall faces: use in the code, boundary conditions, and the geometry prep step

Technical notes

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Contents

1 Summary **1**

2 Definitions and storage **2**
2.1 Geometry quantities per face 2
2.2 Wall faces and the quantity w 3

3 Where P_{bs} and Q_α are used in the equations **3**

4 Boundary conditions on wall faces **4**
4.1 What is implemented 4
4.2 Behaviour in the different regimes 5
4.3 Clipped faces 6

5 The geometry prep step (`b2us_prep`) **6**
5.1 Parameters 6
5.2 Algorithm for `Qalf_mod_style`= 0 7
5.3 Who calls the prep, and the unstructured path 7

6 What reducing the values means physically **7**

7 Practical consequences **8**

8 Tools **8**

9 Recommendations **9**

1 Summary

- `fcPbs` (P_{bs}) is the *magnetic contact area* of a face: the face area S times the fraction of the field direction that crosses it. Everything that moves particles, energy or current *along* **B** through a face is multiplied by P_{bs}. At a wall face it is the size of the opening through which plasma can leave along field lines.

- `fcQalf` is $(\cos, \sin)$ of the poloidal angle α between **B** and the face normal. The two quantities are tied together by

$$w \equiv \frac{P_{bs}}{S} = \left| Q_\alpha^{(0)} \, \frac{B_x}{B} \right| = \cos\alpha \, \frac{B_{pol}}{B} \ , \tag{1}$$

  where B_x/B is `fcBb(0)/fcBb(3)`. On an unmodified `b2fgmtry` this relation holds to machine accuracy on every face (checked numerically).

- Wall faces with small w are called *grazing*. Two scalars stored in `b2fgmtry`, `Qalfmin` and `Qalfmax` (default 10^{-3} each), control how such faces are treated, both by the geometry prep step and by the boundary conditions at run time.
- In the boundary conditions a face with $w < Q_{\alpha,\min}$ is *not* a sheath face: the sheath sinks are replaced by a fixed floor $\propto S\,Q_{\alpha,\max}$, and the guard-cell velocity and potential are pinned to the interior values.
- Reducing P_{bs} (`b2us_prep_Pbs_reduce_fac`) closes the parallel outlet of a grazing face in the bulk equations; raising faces to $w = Q_{\alpha,\max}$ gives faces just above $Q_{\alpha,\min}$ a larger sink.
- Faces that cannot reach $Q_{\alpha,\max}$ because $B_{pol}/B < Q_{\alpha,\max}$ ("clipped" faces) end up inconsistent: $P_{bs}/S = Q_{\alpha,\max}$ but $|Q_\alpha^{(0)} B_x/B| = B_x/B$. Different boundary conditions then see different sinks (Section 4.3).
- The `b2us_prep_*` parameters are *only* read when the prep step runs. On the unstructured (traduit) path of `b2ag` they are ignored ("set but not read" in `b2ag.log`); a Python port (Section 8) applies the wall-face part directly to `b2fgmtry`.

Status of the statements. Sections 2–5 are read from the source (file and line numbers in the Appendix). Section 6 is interpretation. Nothing here was verified by a dedicated test of the Fortran; where a statement is an expectation rather than something read from the code, it is marked.

2 Definitions and storage

2.1 Geometry quantities per face

Face f separates cells `fcCv(f,1)` and `fcCv(f,2)`. A Fortran array `a(nFc,0:1)` is stored component-major in `b2fgmtry`: the first `nFc` values are component 0, the next `nFc` component 1 (`reshape(2,nFc)` in Python).

Array	Meaning
`fcS`	face area S
`fcBb(f,0:3)`	field components at the face; B_x =`fcBb(.,0)`, B =`fcBb(.,3)`
`fcQalf(f,0:1)`	$(\cos,\sin)$ of the angle α between $\mathbf{B}$ and the face normal (poloidal plane)
`fcQbet(f,0:1)`	$(\cos,\sin)$ of β, between the *cell connector* (line joining the two cell centres) and $\mathbf{B}$
`fcQgam(f,0:1)`	$(\cos,\sin)$ of γ, between the connector and the face normal
`fcPbs(f)`	$P_{bs} = S\,Q_\alpha^{(0)} B_x/B$: "(magnetic field pitch)×(surface area), namely the parallel contact area" (comment in `b2agdr.F`)
`fcPbshz`	$P_{bs}\,h_z$; not stored for unstructured files, computed as `fcPbs*fcHz` when the file is read
`Qalfmin`, `Qalfmax`	two scalars stored in `b2fgmtry` (format version $\geq$ 03.002.001)

The three angles are not independent. With $b_z = \pm 1$ the sign of `cvEb(.,2)` (the toroidal field direction),

$$Q_\beta^{(0)} = (Q_\gamma^{(0)} Q_\alpha^{(0)} + Q_\gamma^{(1)} Q_\alpha^{(1)} b_z) b_z, \tag{2}$$

$$Q_\beta^{(1)} = Q_\gamma^{(1)} Q_\alpha^{(0)} - Q_\gamma^{(0)} Q_\alpha^{(1)} b_z. \tag{3}$$

This relation and the unit-vector property of each $(\cos,\sin)$ pair hold to $\sim 10^{-14}$ on every boundary face of an unmodified unstructured file (checked numerically).

2.2 Wall faces and the quantity w

A *wall face* is a face between an interior cell and a *guard* cell (cells `nCi+1..nCv`). Guard faces with $w = 0$ and `fcLbl`≤ 0 belong to the closed-surface (core) boundary; they are never modified by the prep. The vessel-wall faces have `fcLbl`> 0 and

$$w = \frac{P_{bs}}{S} = \left| Q_\alpha^{(0)} \frac{B_x}{B} \right|,$$

the fraction of the field direction that points through the face. A typical wall face has $w \sim 10^{-2}$.

Two different reasons for a small w.

Tangent wall: $|Q_\alpha^{(0)}| \ll 1$ with a normal poloidal pitch B_x/B: the poloidal field runs along the wall.

Low pitch: $|Q_\alpha^{(0)}| \approx 1$ but B_x/B itself is tiny: the field is almost purely toroidal and its poloidal projection barely points at the wall.

The prep logic and the boundary conditions treat both the same way, since both only use w.

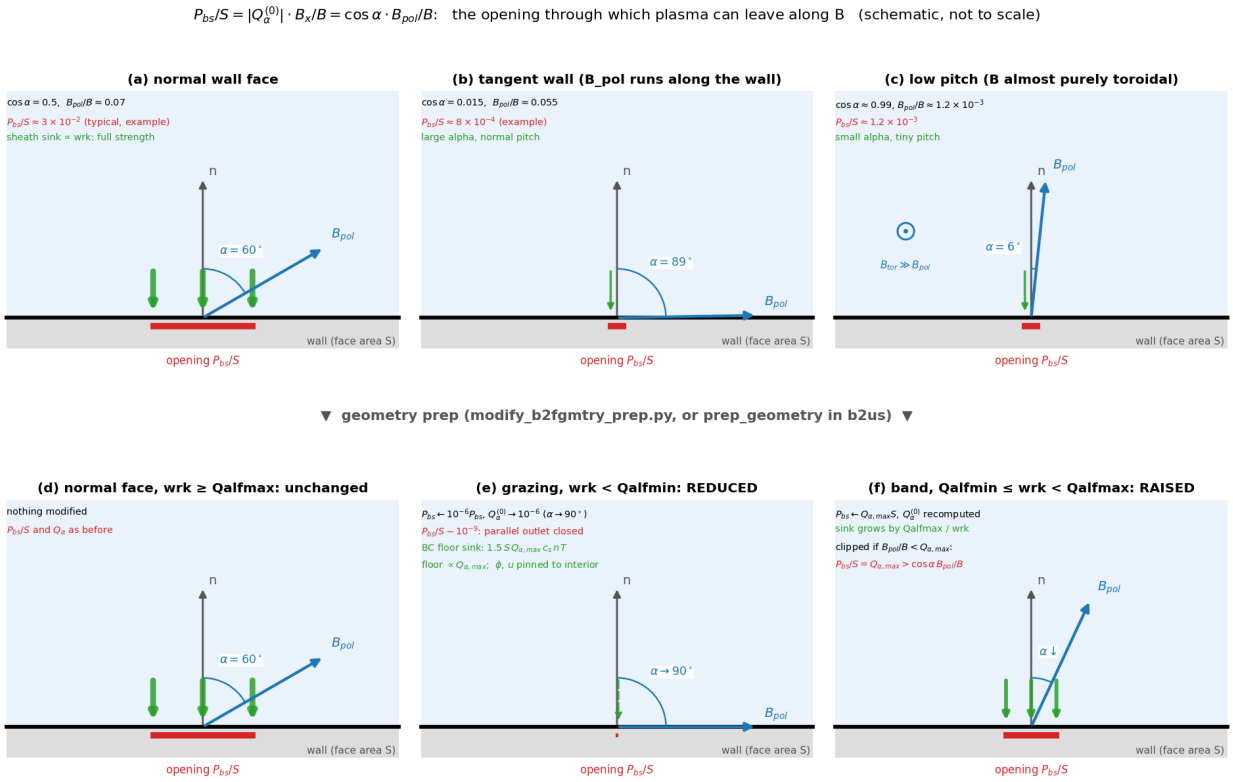


Figure 1: Schematic (not to scale) of one wall face in the poloidal plane: wall (hatched), face normal n, poloidal projection of $\mathbf{B}$ (blue) at angle α to the normal, opening P_{bs}/S (red bar) and the resulting sink (green). Top: (a) normal face, (b) tangent wall, (c) low pitch. Bottom: the effect of the prep step on (d) a normal face (unchanged), (e) a grazing face (reduced), (f) a face in the band $[Q_{\alpha,\min}, Q_{\alpha,\max})$ (raised). Numbers are examples.

3 Where P_{bs} and Q_α are used in the equations

B2.5 evolves the parallel velocity $u_\parallel$; the poloidal projection of a parallel flux through a face is $P_{bs} \times$(flux density).

Routine (line)	Expression	Role
b2tfnb.F:423	flo(:,0)=fcPbs*wrk0+ ...	poloidal particle flux = $P_{bs}\,\overline{u_a}\,n$ plus diffusive and drift parts
gradc_div_par.F:46	wrkf=funf*fcPbs	parallel gradient/divergence of a cell quantity
b2sihs_.F:221	wrkf*fcPbs	$\nabla\cdot(u_e)$ work terms in the energy sources
b2news_.F:440	-fch_p/(qe*(fcPbs+ cutlo))	electron parallel velocity from the parallel current
b2tlhe.F, b2tlhi.F	chvemx=cflme*sqrt(T/ m)*|fcPbs|	free-streaming flux limit: allowed flux $\propto P_{bs}$
b2tlmv.F:78	qfl=cflmv*|fcPbs|*...	parallel viscosity flux limit
b2nxfc.F:88	flub+fcPbshz*wrkf	momentum/convective coefficients
calccoef.F:52--110	weights Qalf, Qbet, Qgam	conductive flux = $\mathrm{con}_0\,\Delta_{\mathrm{cell}} + \mathrm{con}_1\,\Delta_{\mathrm{vertex}}$ with $t_{00}=\cos\alpha$, $t_{10}=\sin\alpha$, $t_{01}\propto\sin\beta$, $t_{11}\propto\cos\beta$
b2txcx.F, b2txcy.F, b2txcc.F	Qalf/Qgam** factors	face conductances contain $\cos\alpha/\cos\gamma$
b2tfhe_.F:134,152	chce(:,0)*fcQalf(: ,0), chce(: ,1)*fcQalf(:,1)	conductivity components weighted with $\cos\alpha$, $\sin\alpha$
b2us_feedback.F:1832	skip face if |Pbs/S| <Qalfmin	feedback loops ignore grazing faces
b2trno.F:377	wrk0>=Qalfmin	with b2trno_set_chcb_0=1: radial coefficients zeroed on wall faces

Wherever P_{bs} is tiny, the cell on the other side of that face is magnetically disconnected from its neighbour (through that face), and every flux limit based on $|P_{bs}|$ is tiny as well.

4 Boundary conditions on wall faces

4.1 What is implemented

The conditions are applied in b2stbc_phys.F as sources in the equation of the *guard* cell. Each source is written $S_0 + S_1\,X$: index 0 the explicit part, index 1 the coefficient multiplying the guard-cell unknown X (n_a, u, T_e, T_i or ϕ). The quantities used below are: S the face area, $w = |Q_\alpha^{(0)} B_x/B|$ (computed from the *stored angles*, not from P_{bs}), $c_s = \sqrt{p_z/\rho_z}$ (for mompar(.,.,2)< 0.5, otherwise $\sqrt{(T_eZ+T_i)/m}$), m_{par} =mompar(.,.,1), γ_e the sheath transmission, and $f = \pm 1$ the face orientation (bcFcOr). The conditions that matter for a wall segment with bccon=14, bcmom=13, bcene=15, bceni=15, bcpot=11 are listed in Table 3. Every one of them tests the *same* criterion, $w < Q_{\alpha,\min}$.

Notes.

- Drift terms (v_{bnd} with vaecrb, vadia, and the *PSch Pfirsch–Schlüter terms) are zero in a run without drifts.
- With bceni_15_style=1 the ion-energy sink is instead $\propto (1+\mathtt{stab_ti})\,t_0\,c_s\,|P_{bs}|$ (line 6123) and there is *no* grazing branch: the $Q_{\alpha,\min}$ test only exists for bceni_15_style=0.
- If conpar(.,.,2)>0, an extra anomalous-diffusion sink $\propto \mathrm{d}n a_0\,S/$conpar2 is added (not discussed here).

Condition (line)	Normal face: $w \geq Q_{\alpha,\min}$	Grazing face: $w < Q_{\alpha,\min}$
particles, `BCCON=14` (1309)	$S_1 = -S\,m_{par}\,c_s\,w$ (more precisely $-S\,m_{par}\max(c_s\lvert w\rvert,\, v_{bc}\operatorname{sgn} w + v_{bnd})$)	$S_0 = 0$, $S_1 = -S\,m_{par}\,c_s\,Q_{\alpha,\max}$
parallel momentum, `BCMOM=13` (3294)	$u_{\text{guard}}\,\lvert P_{bs}h_z\rvert = m_{par}\,v_{bc}\,S\,h_z$, i.e. $u_{\text{guard}} = \pm m_{par}c_s\,\dfrac{S\,w}{\lvert P_{bs}\rvert}$; weight U_{BIG}/ε^2	$u_{\text{guard}} = u_{\text{interior}} -$ `cor9`$(\ldots)$, weight $V_{BIG} = 10^{-2}U_{BIG}$
electron energy, `BCENE=15` style 1 (4496)	$s_1 = \lvert P_{bs}\rvert$	$s_1 = S\,Q_{\alpha,\max}$
	common to both: $t_0 = s_1\,n_e\sqrt{1/2\pi}\,v_{te}\,\exp\!\big(-e(\phi-\phi_{pot})/T_e\big)$; $S_1 = -[(1+\gamma_e+$ `enepar1`$) +$ `stab_te`$]\,t_0$; $S_0 = -[(1-\gamma_e)\,e(\phi-\phi_{pot}) -$ `stab_te`$\,T_e]\,t_0$; zeroed if `b2nppo_restr_po`$\neq 0$ and $S_0 + S_1T_e > 0$	
ion energy, `BCENI=15` style 0 (6061)	$S_1 =$ `enipar1`$\,\sum_s m_{par}\,n_s\,S\,c_s\,w -$ `stab_ti`$\,n_e\,c_s\,S$; $S_0 =$ `+stab_ti`$\,n_e\,c_s\,S\,T_i$	$S_0 = 0$, $S_1 = -1.5\,S\,Q_{\alpha,\max}\,c_s\,n_i$ (no stabilizer)
potential/current, `BCPOT=11` (7586)	current continuity with Sw: $I_i = e\sum Z m_{par} S c_s w\,n_s$, $I_e = e\,n_e\sqrt{1/2\pi}\,v_{te}\,Sw\,\exp(\ldots)$; $S_0 = (1-\gamma_e)I_e - I_i + t'\phi$, $S_1 = -t'$	$\phi_{\text{guard}} = \phi_{\text{interior}} -$ `cor9`$(\ldots)$, weight $P_{BIG}S$ (no sheath current relation)

Table 3: Boundary conditions on a wall face, normal versus grazing. The grazing branch always replaces the true contact fraction w (or P_{bs}/S) by the constant $Q_{\alpha,\max}$ in the sinks, and pins u and ϕ to the interior cell. `cor9` is `b2stbc_BC2_cor9` (default 0): with the default the pinned values are exactly the interior ones.

- The sheath stabilization coefficients add an implicit diagonal +stab × (flux factor) and the matching explicit part +stab × T_{old} × (same factor), so the net source at the current iterate is unchanged. For the ions with style 0 the factor is $n_e c_s S$, *not* P_{bs}.

4.2 Behaviour in the different regimes

Let w denote the angle-based value of the stored file, and P_{bs}/S the stored contact fraction. The cases (a)–(f) refer to Figure 1.

Case	State in the file	What the conditions of Table 3 see
(a), (d) normal, $w \geq Q_{\alpha,\max}$	$P_{bs} = Sw$; unchanged by the prep	normal branch, all rows consistent; momentum gives $u_{\text{guard}} = \pm m_{par}c_s$
(b), (c) before the prep, $w < Q_{\alpha,\min}$	original $P_{bs} = Sw$	grazing branch with the *stored* $Q_{\alpha,\max}$ (default 10^{-3}): floor $Q_{\alpha,\max}S$; u, ϕ pinned; the bulk face fluxes $\propto P_{bs}$ and $Q_\alpha^{(0)}$ are small but not zero
(c) low pitch with $Q_{\alpha,\min} \leq w < Q_{\alpha,\max}$	$P_{bs} = Sw$	normal branch with the small w (sinks $\propto w$); becomes grazing if $Q_{\alpha,\min}$ is raised above w
(e) reduced, $w < Q_{\alpha,\min}$	P_{bs}, $Q_\alpha^{(0)} \to f_{\text{red}}\times$ original ($\alpha \to 90°$)	grazing branch with floor $Q_{\alpha,\max}S$ (the *new* $Q_{\alpha,\max}$ if it was changed); bulk face fluxes $\propto P_{bs}, Q_\alpha^{(0)}$ essentially zero: the parallel outlet is closed

Case	State in the file	What the conditions of Table 3 see
(f) raised, $Q_{\alpha,\min} \le w < Q_{\alpha,\max}$, unclipped	$P_{bs} = Q_{\alpha,\max} S$, $Q_\alpha^{(0)}$ recomputed, $w = Q_{\alpha,\max}$	normal branch with $w = Q_{\alpha,\max}$; all rows consistent; $u_{\text{guard}} = \pm m_{par} c_s$; sink larger by $Q_{\alpha,\max}/w_{\text{old}}$
(f) raised, clipped, $B_x/B < Q_{\alpha,\max}$	$P_{bs} = Q_{\alpha,\max} S$ but $Q_\alpha^{(0)}$ clipped to ± 1, so $w = B_x/B$	see Section 4.3: rows using w see B_x/B, rows using P_{bs} see $Q_{\alpha,\max}$

Continuity of the sink at $Q_{\alpha,\min}$. With the prep applied, a face just above $Q_{\alpha,\min}$ is raised to $w = Q_{\alpha,\max}$, and a face just below gets the floor $Q_{\alpha,\max} S$: both give $\propto Q_{\alpha,\max} S$. If only the scalars are edited in the file without the prep, a face just above $Q_{\alpha,\min}$ keeps its true $w \approx Q_{\alpha,\min}$ while a face just below gets $Q_{\alpha,\max} S$, so the sink jumps by $Q_{\alpha,\max}/Q_{\alpha,\min}$.

4.3 Clipped faces

A face cannot have P_{bs}/S larger than B_x/B (the case $Q_\alpha^{(0)} = 1$). When the prep (or the Python port) raises a face to $P_{bs}/S = Q_{\alpha,\max}$ and $B_x/B < Q_{\alpha,\max}$, $Q_\alpha^{(0)}$ is clipped to 1, so afterwards

$$\frac{P_{bs}}{S} = Q_{\alpha,\max} \quad > \quad \left| Q_\alpha^{(0)} \frac{B_x}{B} \right| = \frac{B_x}{B}.$$

The Fortran prep does exactly the same. The boundary conditions of Table 3 then see different values depending on which quantity they use:

Condition	uses	sees
particles, ion energy (style 0), potential	$w = \lvert Q_\alpha^{(0)} B_x/B \rvert$	B_x/B (no increase)
electron energy	$\lvert P_{bs} \rvert$	$Q_{\alpha,\max}$ (increased)
parallel momentum	both: $u_{\text{guard}} = \pm m_{par} c_s \, Sw/\lvert P_{bs} \rvert$	$\pm m_{par} c_s \, (B_x/B)/Q_{\alpha,\max}$

The momentum condition therefore forces a guard-cell velocity of only $(B_x/B)/Q_{\alpha,\max}$ times the sound speed: *subsonic* (the Bohm criterion is not imposed) on clipped faces. The stored P_{bs} is also used by the bulk fluxes and flux limits. A way to keep a file self-consistent is to cap P_{bs} at $S\, B_x/B$ on such faces (not implemented in the Fortran).

5 The geometry prep step (`b2us_prep`)

5.1 Parameters

Parameter (`b2us_prep_...`)	Default	Meaning
`Qalfmin`	10^{-3}	wall-face threshold in w; stored in `b2fgmtry`
`Qalfmax`	10^{-3}	minimum contact fraction assigned to wall faces / floor sink in the BCs; stored in `b2fgmtry`
`Pbs_reduce_fac`	10^{-3}	factor applied to P_{bs} of faces with $w < Q_{\alpha,\min}$
`mod_hc`	0	recompute `fcHc` on boundary faces (effective cells); recommended with `b2txcx_increase_transp_coefs`> 1
`set_beta_0`	0	set $\beta = 0$ on faces between boundary cells of the same tube; same recommendation
`Qalf_mod_style`	0	0: wall $\iff$ $w \ge Q_{\alpha,\min}$; 1: end-of-tube faces and neighbours

5.2 Algorithm for `Qalf_mod_style`$= 0$

(`b2us_prep.F`, lines 2595–2735.) For every guard face with `fcLbl`> 0, with $b_z = \pm 1$ the sign of `cvEb(guard,2)`:

1. $w = |Q_\alpha^{(0)} B_x/B|$; the face is a *wall face* iff $w \geq Q_{\alpha,\min}$.
2. **Wall face with $w < Q_{\alpha,\max}$:**

$$P_{bs} \leftarrow Q_{\alpha,\max}\, S\, \mathrm{sign}(P_{bs}),$$
$$Q_\alpha^{(0)} \leftarrow \mathrm{clip}\left(\frac{P_{bs}}{S}\frac{B}{B_x}, -1, 1\right), \quad Q_\alpha^{(1)} \leftarrow \mathrm{sign}(Q_\alpha^{(1)})\sqrt{1-(Q_\alpha^{(0)})^2}.$$

3. **Every wall face:** $Q_\beta^{(0)} = \pm b_z$ (sign of $Q_\alpha^{(0)}$), $Q_\beta^{(1)} = 0$, $Q_\gamma^{(0)} = |Q_\alpha^{(0)}|$, $Q_\gamma^{(1)} = \pm Q_\alpha^{(1)} b_z$. That is, $\beta = 0$ and $\gamma = \alpha$: the connector from interior to guard cell is declared parallel to **B**.
4. **Grazing face ($w < Q_{\alpha,\min}$):** $P_{bs} \leftarrow f_{\mathrm{red}} P_{bs}$, Q_α recomputed from the new P_{bs} as in step 2, and Q_β recomputed from the unchanged Q_γ with the relation of Section 2.

Optionally `mod_hc` rescales `fcHc` on the boundary faces and `set_beta_0` sets $\beta = 0$ on tube faces next to the boundary.

Meaning of step 3. In `calccoef` the flux through a face is $\mathrm{con}_0\,\Delta_{\mathrm{cell}} + \mathrm{con}_1\,\Delta_{\mathrm{vertex}}$ with weights $\cos\alpha$, $\sin\alpha$ on the cell difference and $\propto \sin\beta$, $\cos\beta$ on the tangential vertex difference. With $\beta = 0$ the large parallel conductivity no longer couples to the tangential difference, and the factor $\cos\alpha/\cos\gamma$ in the conductances is 1. For faces whose guard cell was already built with the connector along **B** ($\beta \approx 0$) this changes nothing; for faces with a connector at a large angle to **B** it changes the conduction through the face substantially, in a direction that depends on the geometry. This part of the prep is a separate physics choice from the Q_α/P_{bs} treatment.

5.3 Who calls the prep, and the unstructured path

`prep_geometry` is called from `preprocessing/b2agdr_st.F:625` (structured grid converted inside `b2ag`) and `convert/b2us.F:186`. The unstructured driver `b2agdr.F` (traduit input, `*formatunstructured` in `b2ag.dat`) calls only `b2agfs`, `b2agmt` and, if `b2agdr_redef_gmtry`$\neq 0$, `b2agbs`; it *never* calls the prep. Hence in `b2ag.log`:

```
*** IPMAIN--WARNING: parameter has been set but not read: b2us_prep_qalfmin
*** IPMAIN--WARNING: parameter has been set but not read: b2us_prep_qalfmax
... (mod_hc, pbs_reduce_fac, set_beta_0 likewise)
```

At run time `b2us_geo.F:345--354` reads `Qalfmin`/`Qalfmax` from `b2fgmtry` *after* the input parameter, so the value in the file wins. A file produced by the unstructured `b2ag` therefore carries the defaults $10^{-3}/10^{-3}$ regardless of `b2ag.dat`. Note also that `b2agdr_redef_gmtry'0'` disables the flux-constancy correction `b2agbs`, which is on by default since 3.0.6.

6 What reducing the values means physically

- $P_{bs} \to 0$ **on a face** means: no field line crosses it in the poloidal projection, or, in B2.5's 2D picture, the flux tube does not end on that piece of wall. Parallel convection, parallel conduction and the sheath sink through that face vanish in the bulk equations.
- **Reducing by** f_{red} is the numerical way of saying "this is not a target". The face keeps a nonzero value only to avoid divisions by zero (`fcPbs+cutlo`).

- $Q_{\alpha,\max}$ **as a floor** (grazing faces): a regularised, small, non-physical sink $\propto S\,Q_{\alpha,\max}\,c_s\,n$ representing, loosely, "some plasma still reaches the wall" (cross-field transport, field-line tilt below the resolution of the 2D grid). Its size is a modelling choice.

- $Q_{\alpha,\min}$ is where the code stops trusting the stored angle: below it the wall is not a sheath; above it, the sheath applies with the true projection.

- **Insulating potential:** on grazing faces the guard potential is pinned to its interior neighbour, so the sheath current relation is not applied there.

For a flux tube that ends on a face with tiny w, the end cell has (i) an almost closed parallel outlet, (ii) a flux limit $\propto |P_{bs}|$ that is tiny, and (iii) at most the fixed floor sink. Plasma arriving along the tube must leave radially or accumulate in the end cell ("dead end"); this is the situation to look for when density, temperature or parallel velocity pile up at the end of a flux tube that touches the wall at a shallow angle.

7 Practical consequences

- **Where the effect is largest.** The sinks of the normal branch scale with w, so the strongest effect is on the faces with the smallest w; the floor $Q_{\alpha,\max}S$ is the only sink left on grazing faces. Raising $Q_{\alpha,\max}$ strengthens the floor on every grazing face in proportion, and strengthens the raised faces by $Q_{\alpha,\max}/w_{\text{old}}$ (at most $Q_{\alpha,\max}/Q_{\alpha,\min}$).

- **Energy balance of an end cell.** Whether heat still accumulates at a grazing contact is a balance between the new sink and whatever source or flux feeds the end cell. A useful check is the net ion/electron heat leaving the end cell through each of its faces (from the face fluxes in the plasmastate), before and after the change.

- **Potential and current.** Grazing faces have no sheath current relation; if current has no other sink the potential in adjacent cells can drift. Faces just above $Q_{\alpha,\min}$ have a sheath current capacity $\propto w$, i.e. very small.

- **The potential cap.** The ratio limiter `b2nppo_restr_po` limits $|e\phi| \leq$ `restr_po` $\times\, T_e$ locally. In cold cells (low T_e or at the temperature floor) the cap is much smaller than the potential imposed by the neighbours, so a tight cap can conflict with charge conservation. This interacts with any change of the wall treatment and should be varied separately.

- **Change one thing at a time.** Geometry (Qalf/Pbs), angle overwrite ($\beta = 0$), floors/caps and the start state each change the wall problem; compare short start-up runs from the same state in separate run directories.

8 Tools

Script	Purpose
`modify_b2fgmtry_prep.py`	Python port of the wall-face part of the prep for unstructured `b2fgmtry`: `--qmin--qmax--pbs-reduce`. Checks a byte-exact read/write round trip first, never overwrites the input unless `--inplace` (original kept as `.orig`), refuses `-o` equal to the input. `--angleskeep` (default) leaves Q_γ alone and recomputes Q_β only on raised faces; `--anglesprep` also overwrites Q_β, Q_γ on all wall faces. `--scalars-only` stores just `Qalfmin`/`Qalfmax`. `--zboxZMINZMAX` / `--boxRMINRMAXZMINZMAX` (repeatable) restrict the modification to wall faces whose midpoint lies in the box; all other faces keep their original P_{bs} and angles. The scalars remain global (see below). `--dry-run` prints the summary only. Not ported: `mod_hc`, `set_beta_0`, `Qalf_mod_style=1`.
`plot_qalf_wall.py`	Maps the wall faces. Default: classes by w (core-type, grazing, band, normal). `--refORIG`: classes from the original file, coloured by whether the prepared file was actually reduced/raised. `--showpbs`: stored $\log_{10}(\lvert P_{bs}\rvert/S)$. `--showdpbs` (with `--ref`): $\log_{10}$ of the ratio to the original. `--compareQ2` circles faces whose grazing status differs for another $Q_{\alpha,\min}$; `--listN` prints the faces.
`sketch_qalf_pbs.py`	Regenerates the schematic of Figure 1.

The scalars are global. `Qalfmin` and `Qalfmax` are single numbers in `b2fgmtry`. Restricting the geometry change to a region therefore does not restrict the boundary conditions: every wall face with $w < Q_{\alpha,\min}$ anywhere, modified or not, uses the floor $Q_{\alpha,\max}S$. Raising $Q_{\alpha,\max}$ for one region raises the floor on all grazing faces; the script lists the unmodified faces whose sink changes for this reason. A region-dependent floor can only be obtained through the geometry (for example by moving a face out of the grazing branch), not through the scalars.

Reading the plots. Without `--ref` the plot shows the file as the code will see it (the classification of the boundary conditions uses the angle-based w). With `--ref` the plot shows what was done to the file compared with the original; the two differ on raised faces, which move to $w = Q_{\alpha,\max}$, and on clipped faces, whose angle-based w stays below $Q_{\alpha,\max}$.

9 Recommendations

1. Always keep the unmodified `b2fgmtry` and generate variants from it into new file names; the prep is not idempotent (reducing twice multiplies P_{bs} by f_{red}^2).
2. Keep $Q_{\alpha,\max} \geq Q_{\alpha,\min}$ and, if editing only the scalars, prefer equal values to avoid the jump in the sink at $Q_{\alpha,\min}$.
3. Decide explicitly whether to overwrite the connector angles ($\beta = 0$, $\gamma = \alpha$); on a grid built with the connector along $\mathbf{B}$ the overwrite changes little, otherwise it is a separate physics choice.
4. Inspect clipped faces ($B_x/B < Q_{\alpha,\max}$): either accept the inconsistency or cap P_{bs} at $S\,B_x/B$.
5. Remove or comment the unused `b2us_prep_*` lines in `b2ag.dat` on the unstructured path so the log does not suggest they are active.
6. If a parameter set comes from a case built through the structured → `b2us` route, ask how the geometry was produced: the prep step only exists there.

Appendix: source locations

File	Content
`modules/b2us_geo.F:226-227`	defaults $Q_{\alpha,\min} = Q_{\alpha,\max} = 10^{-3}$
`modules/b2us_geo.F:345-354,404-407`	read (file overrides input) and write of the scalars
`modules/b2us_geo.F:954`	`fcPbshz=fcPbs*fcHz` at file read
`modules/b2us_prep.F:2494-2503`	reads of the `b2us_prep_*` parameters
`modules/b2us_prep.F:2595-2735`	wall-face classification, raising, reduction, `mod_hc` per face
`modules/b2us_prep.F:2741-2803`	boundary ordering, `cvAlongBoundary`, `set_beta_0`
`modules/b2us_prep.F:2909ff`	`mod_hc` effective-cell recalculation
`sources/b2stbc_phys.F:605-643`	`csbc_is`, `vbc_is`: c_s, w, v_{bnd}, v_{bc}
`sources/b2stbc_phys.F` 1309, 3294, 4496, 6061, 6123, 7586	BC branches (Table 3)
`sources/b2stbc_phys.F:238-248`	ε, U_{BIG}, V_{BIG}, P_{BIG}, `cor9`
`utility/calccoef.F:52-110`	use of Q_α, Q_β in the face flux
`transport/b2trno.F:377`	`set_chcb_0` use of $Q_{\alpha,\min}$
`modules/b2us_feedback.F:1832`	feedback skips grazing faces
`preprocessing/b2agdr.F`, `b2agdr_st.F:625`, `convert/b2us.F:186`	who calls the prep